

CENTRAL UNIVERSITY OF KASHMIR

Tulmulla, Ganderbal, Jammu and Kashmir, 191131.



Masters in Technology (M. Tech) Computer Science & Engineering (CS&E)

**COURSE STRUCTURE
NEP-2020
[BoS – 18/12/2024]**

School of Engineering & Technology
Department of Information Technology

Course Structure for M.Tech. Computer Science & Engineering

Semester I					
S. No.	Course Code	Course Title	Credits	CIA	ESE
1	MTCS 101	Object Oriented Software Engineering	4	50	50
2	MTCS 102	Algorithms & Algorithmic Complexity	4	50	50
3	MTCS 103	High-Performance Computer Architecture	4	50	50
4	MTCS 104 L	Algorithms & Algorithmic Complexity Lab	2	50	50
5	MTCS 105 L	High-Performance Computer Architecture Lab	2	50	50
6	AEC-I	Ability Enhancement Course-I	2	50	50
7	VAC-I	Value Added Course-I / Elective	2	50	50
		Total	20		
Semester II					
S. No.	Course Code	Course Title	Credits		
1	MTCS 201	Next Generation Networks	4	50	50
2	MTCS 203	Data Science	4	50	50
3	MTCS 204 L	Next Generation Networks Lab	2	50	50
4	MTCS 205 L	Data Science Lab	2	50	50
5	MTCS E	Elective /SEC-I /(MDC-I)	4	50	50
6	AEC-II	Ability Enhancement Course -II	2	50	50
7	VAC –II / Elective	Value Added Course –II / Elective	2	50	50
		Total	20		

Course Structure for M.Tech. Computer Science & Engineering

Semester III			
S. No.	Course Code	Course Title	Credits
1	MTCS 302	Network Security	4
2	MTCS 303	Research Project Phase I	4
3	MTCS 304	Network Security Lab	2
4	MTCS 305	Research Methodology	2
5	MTCS 306	Optimisation Techniques	4
6	MTCS E***	Elective II/(MDC-II)	4
		Total	20

Semester IV			
S. No.	Course Code	Course Title	Credits
1	MTCS 401	Research project Phase II (Dissertation)	20
		Total	20

Total Credits	80
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Course Structure for M.Tech. Computer Science & Engineering

List of Electives			
S. No.	Course Code	Course Title	Credits
1	MTCS E001	Compiler Design	4
2	MTCS E002	Natural Language Processing	4
3	MTCS E003	Multimedia Technologies	4
4	MTCS E005	VLSI	4
5	MTCS E006	Software Project Management	4
6	MTCS E007	Expert Systems	4
7	MTCS E008	Advanced Java & Android Programming	4
8	MTCS E009	Computer Graphics	4
9	MTCS E010	Embedded Systems	4
10	MTCS E011	Elective on Special Topics	4
11	MTCS E012	Cloud Computing	4
12	MTCS E013	Design Practices	4
13	MTCS E014	Advanced Object Oriented Programming System (OOPs)	4
14	MTCS E015	Image Processing	4
15	MTCS E016	System Analysis & Design	4
16	MTCS E017	e-Commerce	4
17	MTCS E018	Operation Research	4
18	MTCS E019	Cyber Forensics	4
19	MTCS E020	Multicore Architecture & Programming	4
20	MTCS E021	Artificial Intelligence	4
21	MTCS E023	Artificial Neural Network	4
22	MTCS E024	Real-Time Operating System	4
23	MTCS E025	Logic Programming	4
24	MTCS E026	Advanced Graph Theory	4
25	MTCS E027	Advanced Engineering Mathematics	4
26	MTCS E028	Formal language Automata Theory	4
27	MTCS E029	Modeling & Simulation	4
28	MTCS E030	Advanced Computer Vision	4
29	MTCS E031	Distributed Computing	4
30	MTCS E032	BioInformatics	4
31	MTCS E033	Computer Based Numerical Techniques	4

Course Structure for M.Tech. Computer Science & Engineering

List of Electives			
S. No.	Course Code	Course Title	Credits
32	MTCS E034	Enterprise Resource Planning (ERP)	4
33	MTCS E035	Reconfigurable Computing	4
34	MTCS E036	Management Information Systems	4
35	MTCS E037	Advanced Microprocessor	4
36	MTCS E038	Web Technology	4
37	MTCS E039	Big Data	4
38	MTCS E040	Advanced RISC Machine (ARM)	4
39	MTCS E041	Machine Learning	4
40	MTCS E042	Blockchain Technology	4
41	MTCS E043	Deep Learning	4
42	MTCS E044	Python for Data Analysis , AI and development	4
43	MTCS E045	Quantitative techniques -I	4
44	MTCS E 046	Quantitative Techniques -II	4
45	MTCS E047	Wireless Communication	4
	MTCS ***	SWAYAM MOOC Course (<i>Department Approved</i>)	4

******* In case MOOC course/s is/are equivalent (in credit and syllabus) to the core course offered by the department in other semesters also, and the student is willing to opt for the same. Students can opt for such course/s only if approved by the Department, provided the credit limit for opting for the MOOC courses is maintained.

Multidisciplinary Course basket Semester 1st

S.No	Course Name	Course Code	No. of Credits	Course Pre-Requisites
1.	Computer Programming-I	MDC-02-114-CP	3	Ni/
2.	Computer Programming -II	MDC-02-115-CP	3	MDC-02-114-CP
3.				
4.				
5.				

Value Added Courses,

S.No	Course Name	Course Code	Credit
1.	Cyber Security Awareness	VAC-02-201-CSA	2
2.	Green Computing	VAC-02-201-GC	2
3.			

Elective Courses removed/upgraded

S. No	Course Code	Course Name	Credits	Remarks
1	MTCS E004	Optimisation Techniques	4	Shifted to core group
2				

Programme:	M.Tech. CSE	Semester:	1st	
Course Code:	MTCS 101	Course Title:	Object-Oriented Software Engineering	
Scheme of Instruction			Scheme of Evaluation	
Total Duration	60 Hrs.		Max. Marks	100
Hours/Week	4 Hrs.		Assignment/IA	45
Credits	4		Attendance	05
Instruction Delivery	L:4, T:0, P:0		End Semester Exam	50
Course Basket	Major		Exam Duration	2 Hr.
Instruction Mode	Regular		Evaluation	Grade
Course Objectives	• Develop models using the UML notation. • Apply an iterative, agile process for software development. • Analyze requirements with use cases and create domain models. • Relate analysis and design artifacts to implementation. • Apply object-oriented design principles and patterns. • Understand architectural layers and software engineering project execution. • Document and present project deliverables. • Use an advanced CASE tool for design and development			
Course Outcomes	• Understand various software development process models. • Apply object-oriented principles in software design and implementation. • Analyze and model software requirements using UML. • Use industry-standard software design patterns for effective development. • Develop, test, and deploy object-oriented software solutions efficiently			
Content				
Unit -1	Module – 1	Fundamentals of Software Engineering Introduction to Software Engineering Definition, nature, and importance,Software engineering as a branch of engineering.		
	Module – 2	Software Development Life Cycle (SDLC) Waterfall,V-Shaped,Prototyping,Incremental,Evolutionary, Spiral, Agile models,Stakeholders in software projects.		

	Module – 3	Software Engineering as a Profession Engineering principles in software development,Ethical and legal concerns.
	Module – 4	Stakeholders in Software Development: Roles of developers, testers, clients, and users,Collaboration in software projects
	Module – 5	Software Project Management: Planning, scheduling, tracking Risk management and mitigation
Unit -2	Module – 1	Object-Oriented Concepts & UML Basics of Object Orientation:Classes and objects,Encapsulation, inheritance, polymorphism,Abstraction.
	Module – 2	Unified Modeling Language (UML) Introduction to UML,Importance and applications
	Module – 3	Structural Modeling in UML Class diagrams,Object diagrams
	Module – 4	Behavioral Modeling in UML Use case diagrams,Sequence and activity diagrams
	Module – 5	Architectural Modeling Component diagrams,Deployment diagrams

Unit -3	Module – 1	Object-Oriented Analysis & Design Introduction to Object-Oriented Analysis (OOA) Fundamentals and significance,Developing domain models
	Module – 2	Object-Oriented Design (OOD) Principles SOLID principles,Design patterns
	Module – 3	Software Architecture in OOD Model-driven architecture,Layered architectures
	Module – 4	Object Oriented Design Generic components of OO Design model , System Design process , Partitioning the analysis model , Concurrency and subsystem allocation .
	Module – 5	Object Oriented Design Task Management component, Data Management component
Unit -4	Module – 1	Agile and Iterative Development in OOSE Iterative Development Approaches Introduction to iterative models,Benefits and challenges
	Module – 2	The Unified Process (UP) Inception, elaboration, construction, transition phases,UP disciplines
	Module – 3	Agile Software Development Principles of Agile methodologies Scrum, XP, Kanban
	Module – 4	Object-Oriented Metrics Measurement techniques,Performance evaluation

	Module – 5	Object-Oriented Testing overview of Testing and object oriented Testing strategies.
Note:	Each Unit is Equivalent to 1 Credit – 15 Hrs/ Week of Theory and 1 Credit consist of 5 Modules. Each Module is Equivalent to 3 Hrs or Multiple of 3 Hrs	
Examination	Part – A: Multiple Choice / Short Answer Type based on Level of Study Part – B: Short Type Questions. Part – C: Essay Type Questions. Note: Part – A & B Questions Covers all Units and Part – C from last 2 Units	
Text Book	1. Timothy C. Lethbridge & Robert Langaniere – <i>Object-Oriented Software Engineering</i> 2. Ivar Jacobson – <i>Object-Oriented Software Engineering: A Use Case Driven Approach</i> 3. Roger S. Pressman – <i>Software Engineering: A Practitioner's Approach</i> 4. Craig Larman – <i>Applying UML and Patterns: An Introduction to Object-Oriented Analysis & Design</i>	
References	1. Ali Bahrami – <i>Object-Oriented Systems Development</i> 2. Grady Booch, James Rumbaugh, Ivar Jacobson – <i>The Unified Modeling Language User Guide</i> 3. Fairley R. – <i>Software Engineering Concepts</i> 4. Online Resources & CASE tool documentation	

Programme:	M. Tech. CSE	Semester	1 st	
Course Code:	MTCS - 102	Course Title:	Algorithms & Algorithmic Complexity	
Scheme of Instruction			Scheme of Evaluation	
Total Duration	60 Hrs.		Max. Marks	100
Hours/Week	4 Hrs.		Assignment/IA	45
Credits	4		Attendance	05
Instruction Delivery	4		Seminar/ End Exam	50
Course Type	Major		Exam Duration	2 Hr
Instruction Mode	Regular		Evaluation	Grade
Course Objectives	• The aim of this Course is to learn how to develop efficient algorithms for simple computational tasks and reasoning about the correctness of them. • To analyse the asymptotic performance of algorithms. • To write rigorous correctness proofs for algorithms. • Through the complexity measures, different range of behaviors of algorithms and the notion of tractable and intractable problems will be understood			
Bullet Points Based on Number of Credits (To Depict Knowledge, Skills, Competency, Attitudes and Values to be Attained)				
Course Outcomes	• To apply knowledge of computing and mathematics to algorithm design. • To analyze a problem and identify the computing requirements appropriate for its solution. • To design, implement, and evaluate an algorithm to meet desired needs. • To apply mathematical foundations, algorithmic principles, and computer science theory to the modeling and design of computer-based systems in a way that demonstrates comprehension of the trade-offs involved in design choices. • To apply design and development principles in the construction of software systems of varying complexity. • To function effectively as a member of a team in order to accomplish a common goal.			
Bullet Points Based on Number of Credits (To depict Transferable Knowledge, Skills, Competency, Attitudes and Values to be Achieved)				
Content				
Unit -I	Module – 1	Introduction: Notion of Complexity covering time complexity and		

Unit -II		space complexity. Worst case complexity, Average case complexity
	Module – 2	Big Oh , Big Theta and Big Omega notations with Example and Comparison
	Module – 3	Examples of simple algorithms and illustration of their complexity. Amortized Cost. Iteration and Recursion- Problem solving using iteration and recursion.
	Module – 4	Substitution Method (Forward and Backward) Tree Method, Master Theorem method
	Module – 5	Binary search, Fibonacci numbers, and Hanoi towers and Their Analysis
	Module – 1	List Abstract Data Type: Operations, Use Cases. Array-Based Lists: Insertion, Deletion, Traversal. Pointer-Based Lists: Linked List Basics.
	Module – 2	Stack ADT & Applications: Push, Pop, Infix-Postfix. Queue ADT & Applications: Enqueue, Dequeue, Circular Queue. Implementations (Array vs Linked List): Pros & Cons.
	Module – 3	Dictionary ADT, Open Hashing (Chaining) and Closed Hashing (Linear Probing) with collision Resolution techniques.
	Module – 4	Skip Lists Structure, Binary Tree Basics: Common types and properties of binary trees, Tree Traversals (Pre/In/Post-order),
	Module – 5	Counting of binary trees. Huffman coding using binary trees. Binary search trees: worst case analysis and average case analysis
Unit –III	Module – 1	AVL trees. Splay trees. Priority Queues.
	Module – 2	Binary heaps: insert and delete min operations and analysis. Binomial queues
	Module – 3	Directed Graphs- Data structures for graph representation. Shortest path algorithms: Dijkstra (greedy algorithm) and Bellman-Ford (dynamic programming).
	Module – 4	Depth-first search and Breadth-first search, Directed acyclic graphs, Undirected Graphs
	Module – 5	Minimal spanning trees and algorithms (Prim's and Kruskal) and implementation. All pair Shortest path (Floyd-Warshall), Travelling salesman problem.
Unit- IV	Module – 1	Sorting- Bubble sort, Selection sort, Insertion sort.
	Module – 2	Merge sort, Quicksort, Heap sort.
	Module – 3	Shell sort, Radix sort, Analysis of the sorting methods.
	Module – 4	Selecting the top k elements. Lower bound on sorting.
	Module – 5	Complexity Classes: P class, NP-Complete, NP-Hard, P=NP problem.

Note:	Each Unit is Equivalent to 1 Credit – 15 Hrs/ Week of Theory and 1 Credit consist of 5 Modules. Each Module is Equivalent to 3 Hrs or Multiple of 3 Hrs
Examination	Part – A: Multiple Choice / Short Answer Type based on Level of Study Part – B: Short Type Questions. Part – C: Essay Type Questions. Note: Part – A & B Questions Covers all Units and Part – C from last 2 Units
Text Book	1. T. H. Cormen, C. E. Leiserson, R. L. Rivest, Clifford Stein, “Introduction to Algorithms”, 5th Ed., PHI 2. Ellis Horowitz and SartazSahani, “Computer Algorithms”, Galgotia Publications.
References	1. V. Aho, J. E. Hopcroft, J. D. Ullman, “The Design and Analysis of Computer Algorithms”, Addison Wesley. 2. D. E. Knuth, “The Art of Computer Programming”, 2nd Ed., Addison Wesley. 3. Jon Kleinberg and ÉvaTardos, “Algorithm Design”, 1ST Edition, Pearson. 4. Mark Allen Weiss, “Data Structures and Algorithm Analysis in Java” Addison Wesley. 5. Robert Sedgewick and Kevin Wayne’s, “Algorithms”, Addison-Wesley Professional.

Programme:	M.Tech. CSE	Semester:	1st	
Course Code:	MTCS 103	Course Title:	High Performance Computer Architecture	
Scheme of Instruction			Scheme of Evaluation	
Total Duration	60 Hrs.		Max. Marks	100
Hours/Week	4 Hrs.		Assignment/IA	45
Credits	4		Attendance	05
Instruction Delivery	L:4, T:0, P:0		End Semester Exam	50
Course Basket	Major		Exam Duration	2 Hr.
Instruction Mode	Regular		Evaluation	Grade
Course Objectives	By the end of this course, students will be able to: • Understand the fundamentals of computer architecture, including the software-hardware interface and instruction set design • Analyze memory hierarchy concepts such as cache, virtual memory, MMU, and TLB, and evaluate their impact on system performance. • Explore pipeline architectures, hazard mitigation techniques, and their implementation in modern processors like ARM. • Examine multiprocessor and multicore architectures, focusing on cache coherence, interconnection networks, and multi-threading. • Implement parallel programming techniques using OpenMP, including directives, work-sharing constructs, synchronization, and API usage.			
Course Outcomes	After completing this course, students will be able to: • Demonstrate knowledge of instruction set architectures, addressing modes, and microprogrammed control units in modern processors. • Evaluate memory system performance by analyzing caching techniques, associative memory, and virtual memory management. • Design and optimize pipeline architectures while addressing hazards to improve instruction throughput. • Apply principles of multiprocessor organization, interconnection networks, and coherence protocols in high-performance systems.			

		• Develop parallel applications using OpenMP to enhance computational efficiency on multicore architectures.
Content		
Unit -1	Module – 1	Introduction to Computer Architecture – Basics, software-hardware interface, performance metrics
	Module – 2	Performance Evaluation Techniques – Benchmark programs, performance measurement methods
	Module – 3	Micro-Operations and Instruction Execution – Micro-operations, instruction types, timing and control
	Module – 4	Control Unit Design – Microprogrammed vs. hardware control units, program control principles
	Module – 5	Instruction Set Architecture (ISA) and ARMv8-A – Addressing modes, instruction formats, and ARMv8-A ISA
Unit -2	Module – 1	Memory Technologies and Hierarchy – Auxiliary memory, main memory, memory address mapping
	Module – 2	Cache and Virtual Memory – Associative memory, cache design, virtual memory implementation
	Module – 3	Floating-Point Arithmetic – Representation, addition, subtraction, multiplication, and division
	Module – 4	ARM Cortex-A9 Processor Case Study – MMU and TLB functionality, memory management in Cortex-A9
	Module – 5	Memory Performance and Optimization – Impact of memory design on performance, latency, and bandwidth
Unit -3	Module – 1	Pipelining Fundamentals – Instruction and arithmetic pipelining concepts
	Module – 2	Hazards in Pipelining – Structural, control, and data hazards; forwarding vs. stalling
	Module – 3	Advanced Processing Techniques – Vector processing and its role in high-performance computing
	Module – 4	I/O Systems and Interfaces – Asynchronous data transfer, priority interrupts, and I/O processors
	Module – 5	Multiprocessing and Multicore Architectures – SMP, multi-threading, chip multiprocessors, and ARM10 pipeline
Unit -4	Module – 1	Cache Coherence in Multiprocessors – Issues, hardware solutions, and MESI protocol
	Module – 2	Interconnection Networks – Multistage Cube and Omega Network architectures
	Module – 3	Introduction to OpenMP – Basics, parallel constructs, and directive-based parallelism

	Module – 4	Work-Sharing and Synchronization in OpenMP – Data environment constructs and synchronization techniques
	Module – 5	High-Performance Computing with OpenMP – API libraries for finer control and real-world parallel computing applications
Note:	Each Unit is Equivalent to 1 Credit – 15 Hrs/ Week of Theory and 1 Credit consist of 5 Modules. Each Module is Equivalent to 3 Hrs or Multiple of 3 Hrs	
Examination	Part – A: Multiple Choice / Short Answer Type based on Level of Study Part – B: Short Type Questions. Part – C: Essay Type Questions. Note: Part – A & B Questions Covers all Units and Part – C from last 2 Units	
Text Book	[1] W. Stallings, <i>Computer Organisation & Architecture: Designing for Performance</i>, 9th ed. Prentice Hall International, 2012. [2] M. Mano, <i>Computer System Architecture</i>, 3rd ed. Prentice-Hall, 2001.	
References	[1] D. A. Patterson and J. L. Hennessy, <i>Computer Organization and Design</i>, 2nd ed. Morgan Kaufmann, 1997. [2] C. Hamacher, Z. Vranesic, and S. Zaky, <i>Computer Organization</i>, 5th ed. McGraw-Hill, 2002. [3] R. Chandra, R. Menon, and L. Dagum, <i>Parallel Programming in OpenMP</i>. Morgan Kaufmann, 2001 [4] D. Harris and S. L. Harris, <i>Digital Design and Computer Architecture, ARM Edition</i>. Morgan Kaufmann, 2015. [5] Y. Wang and F. A. Briggs, <i>Computer Architecture and Organization</i>. McGraw-Hill, 1984.	

